COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 031	**SECTION:**	CPE21S2
DATE PERFORMED:	08/05/26	**DATE SUBMITTED:**	08/05/26
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ACTIVITY NO. _4.1__ STACKS

1. PROCEDURE OUTPUT

ILOB.2:

HOA 4.1 > ILOB2.cpp > main()

```cpp
#include <iostream>
#include "stackList.h"

int main() {

    std::cout << "--- PUSHING 'c' 'p' 'e' ONTO THE STACK ---\n";
    //push
    push('c');
    push('p');
    push('e');

    //display the current top
    Top<char>(); // Top should be 'e'
    displayALL<char>();
    std::cout << "\n ----- Popping all the elements -----\n";

    while(!isEmpty<char>()){
        std::cout << " Popped: " << pop<char>() << std::endl;
    }

    std::cout << "\n ------ Testing the Underflow ----- \n";
    pop<char>();

    return 0;
}
```

ARRAYSTACK_.cpp | LINKEDLISTSTACK_.cpp | stacklist.h

HOA 4.1 > C stacklist.h > ...

```cpp
#ifndef STACK_LIST_H
#define STACK_LIST_H

//class node that contains data and the next
template <typename T>
class Node{
    public:
        T data;     // VALUE THAT WE NEED STORE
        Node* next; // POINTER TO THE NEXT NODE
};

//global pointer for the top of the stack
template <typename T>
Node<T>* head = nullptr;

//operations for the stack

// push
template <typename T>
void push(T newData){
    //create a new node and assign the newData
    Node<T>* newNode = new Node<T>;
    newNode->data = newData;
    //new node points to current top, then becomes the new head
    newNode->next = head<T>;
    head<T> = newNode;
}

// pop
template<typename T>
T pop(){
    //check if the stack is empty:
    if(head<T> == nullptr){
        std::cout<< "stack overflow. "<<std::endl;
    }

    //save the current top node
    Node<T>* temp = head<T>;
    //store its data
    T tempVal = temp->data;

    // move the head to the next node
    head<T> = head<T>->next;
    //free the removed node
    delete temp;
    //return the removed value
    return tempVal;
}

// isEmpty
template<typename T>
bool isEmpty(){
    return head <T> == nullptr;
}
```

```
HOA 4.1 > C stacklist.h > ...
  1  #ifndef STACK_LIST_H
 57  template<typename T>
 58  void Top(){
 59      // error catching:
 60      if(isEmpty<T>()){
 61          std::cout<<"Stack underflow "<<std::endl;
 62          return;
 63      }
 64      // display the Top node
 65      std::cout<<"Top of Stack: "<<head<T>->data <<std::endl;
 66  }
 67
 68  //display All
 69  template <typename T>
 70  void displayALL(){
 71      if (head<T> == nullptr){
 72          std::cout << "stack undeflow\n";
 73          return;
 74          }
 75      std::cout<< "Displaying the stack: \n";
 76      Node<T>* temp = head<T>;
 77      while(temp != nullptr){
 78          std::cout << temp->data << "\n";
 79          temp = temp->next;
 80      }
 81      std::cout << "\n";
 82  }
 83  #endif
```

```
PS C:\Users\LENOVO\Downloads\Macasaet_singlyLL\HOA 4.1> cd "c:
$?) { g++ ILOB2.cpp -o ILOB2 } ; if ($?) { .\ILOB2 }
--- PUSHING 'c' 'p' 'e' ONTO THE STACK ---
Top of Stack: e
Displaying the stack:
e
p
c


 ----- Popping all the elements -----
 Popped: e
 Popped: p
 Popped: c

 ------ Testing the Underflow -----
stack overflow.
PS C:\Users\LENOVO\Downloads\Macasaet_singlyLL\HOA 4.1>
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

STACKS USING ARRAYS:

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis

(A+B)+(C-D)	Y	ARRAY STACK: PS C:\Users\LENOVO\Downloads\Macasaet_singly $?) { g++ ARRAYSTACK_.cpp -o ARRAYSTACK_ } ; Enter a String: (A+B)+(C-D) (and) are matching (and) are matching The expression is BALANCED PS C:\Users\LENOVO\Downloads\Macasaet_singly LINKED LIST STACK: PS C:\Users\LENOVO\Downloads\Macasa $?) { g++ LINKEDLISTSTACK_.cpp -o L Enter a String: (A+B)+(C-D) (and) are matching (and) are matching The expression is BALANCED PS C:\Users\LENOVO\Downloads\Macasa	For this the expression is balanced since there is a pair of complete matching parentheses. We pop the opening and match them with their corresponding closing symbol and if they are the same then it is balanced.
((A+B)+(C-D)	N	ARRAY STACK: PS C:\Users\LENOVO\Downloads\Macasaet_sin $?) { g++ ARRAYSTACK_.cpp -o ARRAYSTACK_ Enter a String: ((A+B)+(C-D) (and) are matching (and) are matching ERROR: THE EXPRESSION IS NOT BALANCED! PS C:\Users\LENOVO\Downloads\Macasaet_sin LINKED LIST STACK: PS C:\Users\LENOVO\Downloads\Macasaet_s $?) { g++ LINKEDLISTSTACK_.cpp -o LINKE Enter a String: ((A+B)+(C-D) (and) are matching (and) are matching ERROR: THE EXPRESSION IS NOT BALANCED! PS C:\Users\LENOVO\Downloads\Macasaet_s	It is not balanced since at the end of the program the stack is not empty which means not everything in the stack was popped. The first (does not have any closing parentheses to match it.

((A+B)+[C-D])	Y	ARRAY STACK: PS C:\Users\LENOVO\Downloads\Macasaet_singlyL $?) { g++ ARRAYSTACK_.cpp -o ARRAYSTACK_ } ; Enter a String: ((A+B)+[C-D]) (and) are matching [and] are matching (and) are matching The expression is BALANCED PS C:\Users\LENOVO\Downloads\Macasaet_singlyL LINKED LIST STACK: PS C:\Users\LENOVO\Downloads\Macas $?) { g++ LINKEDLISTSTACK_.cpp -o Enter a String: ((A+B)+[C-D]) (and) are matching [and] are matching (and) are matching The expression is BALANCED PS C:\Users\LENOVO\Downloads\Macas	It is balanced since all the opening symbols have a matching closing symbol. The first is the parentheses containing A+B and the second one is the square brackets containing C-D. The last is the parentheses containing the whole expression.
((A+B]+[C-D]}	N	ARRAY STACK: PS C:\Users\LENOVO\Downloads\Macasaet_singlyLL\HOA 4.1> cd "c $?) { g++ ARRAYSTACK_.cpp -o ARRAYSTACK_ } ; if ($?) { .\ARRA Enter a String: ((A+B]-[C-D]} ERROR: (and] are NOT matching ERROR: THE EXPRESSION IS NOT BALANCED! PS C:\Users\LENOVO\Downloads\Macasaet_singlyLL\HOA 4.1> LINKED LIST STACK: PS C:\Users\LENOVO\Downloads\Macasaet_singlyLL\H $?) { g++ LINKEDLISTSTACK_.cpp -o LINKEDLISTSTAC Enter a String: ((A+B]+[C-D]} ERROR: (and] are NOT matching ERROR: THE EXPRESSION IS NOT BALANCED! PS C:\Users\LENOVO\Downloads\Macasaet_singlyLL\H	It is not balanced since the closing square bracket beside B has no corresponding opening square bracket. And the } at the end also has no corresponding opening symbol. The first two opening parentheses at the start have no corresponding closing parentheses.

Tools Analysis:

How do the different internal representations affect the implementation and usage of the stack?

- Using an array stack means that the stack will have a limited number of elements that it can store. All the code is also seen at the cpp file. On the other hand, using linked list stack means that the cpp file will be shorter since the functions will be put in the header file. The stacks' number of elements are also not fixed. Using this approach also makes the stacks into a linked list whereas the head can be considered the top.

CONCLUSION:

In this activity, I implemented the basics of stacks and also the different approaches to creating stacks. In this, array stacks and linked list stack were used. Both can be used depending on if you know how many elements in a stack you want. For me, the array stacks are much easier to use since in linked list you will have to deal with a lot of pointers.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Debugging**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **If I am really struggling debugging my code, I first try to solve the problem on my own but if it takes too long and gives up, I use AI to help me.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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